

Abrar Bari

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Portfolio

EDUCATION

Texas Tech University

Bachelor of Science in Computer Science | GPA: 4.00/4.00

Minor: Mathematics | Concentration: Artificial Intelligence

Graduated May 2026

Lubbock, TX

TECHNICAL SKILLS

Languages: Python, JavaScript, TypeScript, C, Java, HTML, CSS, SQL

Dev Tools: Docker, AWS, Git, GitHub, VS Code, Tableau, Postman, Jira, Render, Vercel

Frameworks/Libraries: React, Next.js, Node.js, Express, PostgreSQL, MongoDB, Prisma, Redis, BullMQ, TensorFlow, LangChain

EXPERIENCE

Undergraduate Research Assistant

Sept 2025 – May 2026

Dept. of Chemical Engineering, Texas Tech University

Lubbock, TX

- Developed a transfer learning CNN pipeline using ResNet-V2 for binary classification of sickle vs. healthy red blood cells on CTV image sequences, achieving 80% validation accuracy and F1 scores of 82%/80% on a balanced 244-image dataset.
- Engineered an overlapping tiling algorithm for 2048×1536 microscopic images using YOLOv8/v11 object detection, achieving 95% precision and 93% recall across RBC classes while co-authoring a research paper on blood smear analysis.

Undergraduate Teaching Assistant

Aug 2025 – May 2026

Dept. of Computer Science, Texas Tech University

Lubbock, TX

- Facilitated weekly office hours for CS 2413 Data Structures (DS-I), guiding 70+ students through LeetCode-style problem solving across arrays, stacks, queues, linked lists, hash maps, binary trees, and recursion in C, reinforcing algorithmic thinking and debugging skills.
- Evaluated and graded weekly lab assignments, midterms, and class projects while proctoring exams, ensuring academic integrity and delivering timely, constructive feedback that improved student comprehension and performance.

Machine Learning Fellow

Aug 2025 – Dec 2025

AI Studio Fellow @ Meta AI – Break Through Tech, Cornell University

Remote

- Fine-tuned and published BERTForSequenceClassification to Hugging Face Hub on 11K+ Reddit comments for demographic bias detection, achieving 82% accuracy and F1 of 0.84 via hyperparameter grid search across learning rates.
- Devised a fairness evaluation pipeline with per-category subgroup analysis across 5 demographic slices, benchmarking generalization on CrowS-Pairs and BBQ datasets to surface domain shift and bias blind spots.

IT Supervisor

Sept 2023 – Aug 2025

IT Help Central, Texas Tech University

Lubbock, TX

- Supervised and mentored a team of analysts and technicians supporting a 35,000+ student campus, delegating frontline support across Microsoft 365, Canvas, network, hardware, and dorm infrastructure while logging/tracking incidents in ServiceNow, reducing average resolution time by 25%.
- Orchestrated campus-wide incident response workflows in ServiceNow, managing issue queues, coordinating cross-department escalations, and delivering real-time coaching to accelerate ticket resolution accuracy by 15%.

PROJECTS

InterviewCraft - Next.js, TypeScript, Node.js, Express, PostgreSQL, Prisma, Zod

Demo | GitHub

- Built a full-stack AI interview preparation platform using Next.js, Express, and PostgreSQL that generates role-specific interview questions, structured feedback, and guided mock interview workflows for personalized interview practice.
- Implemented production-ready REST APIs with Prisma ORM and Zod validation, enforcing schema-safe request handling through HttpOnly cookies, strict CORS policies, and route-level rate limiting to strengthen backend security, reliability, and maintainability.

PaperPilot - Next.js, Express, Prisma, PostgreSQL 16, pgvector, BullMQ, Redis, OpenAI, Docker, Turbo

GitHub

- Built a production-grade RAG platform using Next.js, Express, PostgreSQL + pgvector, BullMQ, Redis, and OpenAI, improving retrieval quality through hybrid search tuning and scalable asynchronous document processing for accurate research paper answer grounding.
- Developed an asynchronous PDF ingestion and embedding pipeline with BullMQ workers, pdf-parse, OpenAI embeddings, and pgvector indexing, processing full research papers in under 2 minutes with retries, structured logging, request tracing, retrieval benchmarking, and Dockerized backend services for production-ready workflows.

ManageMySubscription - Next.js, TypeScript, Node.js, PostgreSQL, Docker

GitHub

- Engineered a secure full-stack subscription management platform using Next.js, Express, PostgreSQL, and Docker, implementing 11 REST API routes with JWT authentication, bcrypt password hashing, ownership-based access control, and Arcjet token-bucket rate limiting to protect against abuse and unauthorized access.
- Designed a normalized PostgreSQL schema with indexed relational models, SQL-enforced integrity constraints, and an automated 5-day renewal reminder workflow supporting multiple billing frequencies, currencies, and subscription categories.